SAT Math Mastery · Session 7

Data Analysis and Statistics

Time	3 hours: Part A in class (about 2 hours of solving), then Part B, the self-assessment, done independently (about 50 minutes), and a short error log.
Topics	ratios, rates and unit conversion · mean, median and mode from tables · how adding, removing or changing values changes the mean and standard deviation · box plots and spread · sampling and study design · margin of error and estimates · probability and two-way tables · scatterplots, slope and residuals
Name / Date	_____ _____

PART A · IN CLASS — 35 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal). Grey boxes show exactly what to type in Desmos (or a calculator), one expression per line; tap a grey dot to read an intersection or a vertex.

1. HARD · DESMOS

Study room A has x students and study room B has 72 students. One-third of the students in room A move to room B. Then one-fourth of the students who are now in room B move to room A. After both moves, the ratio of the number of students in room A to the number in room B is 5 to 7. How many students were in room A originally?

(A) 72 **(B) 36 ✓**

(C) 45 (D) 63

1. After the first move room A has $\frac{2x}{3}$ students and room B has $72+\frac{x}{3}$. After the second move room A has $\frac{3x}{4}+18$ and room B has $54+\frac{x}{4}$.
2. A ratio of $5:7$ means $7A=5B$. Type `y=7(3x/4+18)` and `y=5(54+x/4)`.
3. Tap the intersection: (36, 315). Its x-value is the original number in room A.
4. Check with $x=36$: room A ends with 45 and room B with 63, and $45:63$ is $5:7$.

Answer: (B)

2. HARD · NO CALCULATOR NEEDED

A tram increases its speed at a constant rate from 54 kilometers per hour to 90 kilometers per hour in 20 seconds. What is the tram's acceleration, in meters per minute squared?

1. Convert to meters per second: $54\text{ km/h}=\frac{54\,000}{3600}=15\text{ m/s}$ and $90\text{ km/h}=25\text{ m/s}$.
2. The speed rises by 10 m/s in 20 s, so the acceleration is $\frac{10}{20}=0.5$ meters per second squared.
3. One minute is 60 seconds, so one second squared is $\frac{1}{3600}$ of a minute squared and the number is multiplied by 3600: $0.5\times3600=1{,}800$.

Answer: 1800

3. Hard · Calculator

A car uses fuel at a rate of 6.0 liters per 100 kilometers on a highway and 10.0 liters per 100 kilometers in the city. On one trip the car travels 240 kilometers on the highway and 80 kilometers in the city. Using 1 mile = 1.6 kilometers and 1 gallon = 3.8 liters, which of the following is closest to the car's fuel efficiency for the whole trip, in miles per gallon?

(A) 87

(B) 6

(C) 30

(D) 34 ✓

1. Fuel used: the highway part uses $240 \times \frac{6.0}{100}$ liters and the city part uses $80 \times \frac{10.0}{100}$ liters. Type `240*6/100+80*10/100`: it returns 22.4 liters.
2. Distance: $240 + 80 = 320$ kilometers. Type `(240+80)/1.6`: it returns 200 miles. Type `22.4/3.8`: it returns 5.8947 gallons.
3. Type `200/(22.4/3.8)`: it returns 33.9286, which is closest to 34 miles per gallon.

Answer: (D)

4. Hard · Calculator

On a map, 1 inch represents 1,500 feet. A rectangular parking lot appears on the map as 2.4 inches by 1.6 inches. Which of the following is the actual area of the lot, in square yards? (1 yard = 3 feet)

(A) 8,640,000

(B) 320,000

(C) 960,000 ✓

(D) 2,880,000

1. Real lengths: Type `2.4*1500`: it returns 3,600 feet and Type `1.6*1500`: it returns 2,400 feet.
2. Change to yards before multiplying: Type `3600/3`: it returns 1,200 yards and Type `2400/3`: it returns 800 yards.
3. Type `1200*800`: it returns 960,000 square yards. (Multiplying in feet first gives square feet, 8,640,000, which is not what was asked.)

Answer: (C)

5. Hard · Calculator

A cyclist rides a 90-kilometer route. She covers the first 60% of the distance at an average speed of 27 kilometers per hour, rests for 30 minutes, and then covers the rest of the route at an average speed of 24 kilometers per hour. What was her average speed for the whole trip, including the rest, in meters per second?

(A) 22.5

(B) 6.25 ✓

(C) 7.08

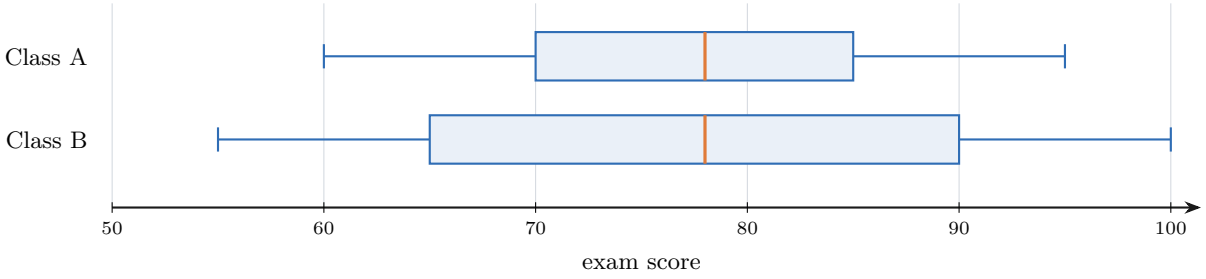
(D) 7.14

1. First part: Type `0.6*90`: it returns 54 km at 27 km/h, so Type `54/27`: it returns 2 hours.
2. Rest of the route: Type `90-54`: it returns 36 km at 24 km/h, so Type `36/24`: it returns 1.5 hours. Total time with the rest: Type `2+1.5+0.5`: it returns 4 hours.
3. Average speed: Type `90/4`: it returns 22.5 km/h. Convert to meters per second: Type `22.5*1000/3600`: it returns 6.25.

Answer: (B)

6. HARD · NO CALCULATOR NEEDED

Box plots summarize final exam scores for two sections of a course.
Class A: minimum 60, $Q1 = 70$, median $= 78$, $Q3 = 85$, maximum 95.
Class B: minimum 55, $Q1 = 65$, median $= 78$, $Q3 = 90$, maximum 100.



Which statement is best supported by these box plots?

(A) The two classes have the same median and the same interquartile range (IQR).

(B) The two classes have the same median, but Class B's scores are more spread out, as shown by its greater IQR. ✓

(C) Class A has a greater median than Class B.

(D) Class B's scores are less spread out than Class A's, as shown by its smaller IQR.

1. The medians are both 78, so the statement that Class A has the greater median is out, and so is any statement that treats the medians as different.
2. The interquartile range is $Q3 - Q1$: Class A has $85 - 70 = 15$ and Class B has $90 - 65 = 25$.
3. The IQR measures the spread of the middle half of the scores, so Class B is more spread out. The statement with the same median and a greater IQR for Class B is the one the box plots support.

Answer: (B)

7. HARD · NO CALCULATOR NEEDED

A data set of 24 different numbers has a mean of 50 and a median of 50. A new data set is created by adding 5 to each number in the original data set that is greater than the median and subtracting 5 from each number that is less than the median. Which of the following measures does NOT have the same value in both the original and new data sets?

(A) Median

(B) Mean

(C) Sum of the numbers

(D) Standard deviation ✓

1. The 24 numbers are all different and the median is 50, so exactly 12 are below 50 and 12 are above 50.
2. Twelve numbers go up by 5 and twelve go down by 5: the sum changes by $12(5) - 12(5) = 0$. The sum and the mean (50) do not change.
3. The two middle values move one up and one down by 5, so their average, the median, is still 50.
4. Every value moves 5 farther from the mean, so the numbers are more spread out: only the standard deviation changes.

Answer: (D)

8. HARD · NO CALCULATOR NEEDED

The mean score of 12 students on a quiz was 16.5 points. If the highest individual score is removed, the mean score of the remaining 11 students becomes 15 points. What was the highest score?

1. Total of all 12 scores: $12 \times 16.5 = 198$.

2. Total of the other 11 scores: $11 \times 15 = 165$.

3. The highest score is the difference: $198 - 165 = 33$.

Answer: 33

9. HARD · NO CALCULATOR NEEDED

A weather station recorded the high temperature, in degrees Fahrenheit, on each of 15 days. The mean of these 15 readings was 42. After the reading from one day was removed as a recording error, the mean of the remaining 14 readings was 40. What was the removed reading?

1. Total of the 15 readings: $15 \times 42 = 630$.

2. Total of the other 14 readings: $14 \times 40 = 560$.

3. The removed reading is the difference: $630 - 560 = 70$ degrees.

Answer: 70

10. HARD · NO CALCULATOR NEEDED

A data set is 10, 14, 18, 22, 36. A new value of 18 is added to the set. Which of the following statements about the new data set is (are) true?

I. The mean is less than the mean of the original set. II. The median is the same as the median of the original set. III. The range is the same as the range of the original set.

(A) I only

(B) I and II only

(C) II and III only

(D) I, II and III ✓

1. Original set: mean $= \dfrac{100}{5} = 20$, median $= 18$, range $= 36 - 10 = 26$.

2. New set $10, 14, 18, 18, 22, 36$: the sum is 118, so the mean is $\dfrac{118}{6} \approx 19.67$. This is less than 20, so I is true.

3. The two middle values are 18 and 18, so the median is still 18. II is true.

4. The smallest and largest values are unchanged, so the range is still 26. III is true: all three statements hold.

Answer: (D)

11. HARD · NO CALCULATOR NEEDED

A group of 30 people has a mean age of 70 years. When a second group of 20 people is added, the mean age of all 50 people is 76 years. What is the mean age, in years, of the second group?

1. Total age of all 50 people: $50 \times 76 = 3{,}800$. Total age of the first group: $30 \times 70 = 2{,}100$.

2. The second group's total is $3{,}800 - 2{,}100 = 1{,}700$.

3. Its mean is $\dfrac{1{,}700}{20} = 85$ years.

Answer: 85

12. HARD · NO CALCULATOR NEEDED

The frequency table shows the scores of 30 students on a 5-point quiz.

Score	1	2	3	4	5
Frequency	3	5	8	9	5

What is the median score?

(A) 2.5

(B) 3 ✓

(C) 3.5

(D) 4

1. There are 30 scores, so the median is the average of the 15th and 16th scores.

2. Running totals of the frequencies: 3, 8, 16, 25, 30. Scores 9 to 16 are all 3, so the 15th and the 16th scores are both 3.

3. The median is $\dfrac{3+3}{2} = 3$.

Answer: (B)

13. HARD · NO CALCULATOR NEEDED

A data set has eight values: 12, 15, 18, 20, 20, 22, 25, 28. The values 12 and 28 are removed and the values 19 and 21 are added in their place. Which of the following describes the mean and the standard deviation (SD) of the new data set compared with the original data set?

(A) Mean smaller, SD smaller

(B) Mean same, SD larger

(C) Mean same, SD smaller ✓

(D) Mean same, SD same

1. The original sum is 160, so the mean is 20. The values removed add to $12+28 = 40$ and the values added are $19+21 = 40$, so the sum and the mean do not change.

2. Distances from the mean 20: the removed values were 8 away on each side. The added values 19 and 21 are only 1 away on each side.

3. The data are now closer to the mean, so the standard deviation gets smaller: mean the same, SD smaller.

Answer: (C)

14. HARD · DESMOS

A data set consists of the five values 5, 8, 17, x, x, where x is a real number. The variance of a data set is the mean of the squared deviations of its values from their mean. What is the least possible value of the variance of this data set?

(A) 26

(B) 78

(C) 15.6 ✓

(D) 10

1. The mean is $\frac{30+2x}{5}$, so the variance is the average of the five squared distances from it.

2. Type `y=((5-(30+2x)/5)^2+(8-(30+2x)/5)^2+(17-(30+2x)/5)^2+2(x-(30+2x)/5)^2)/5`.

3. Tap the lowest point of the curve: (10, 15.6). The least possible variance is the y-value.

4. Check with $x = 10$ (mean 10): Type `((5-10)^2+(8-10)^2+(17-10)^2)/5`: it returns 15.6.

Answer: (C)

15. HARD · NO CALCULATOR NEEDED

A data set has 10 values: 31, 38, 42, 45, 47, 52, 53, 58, 65, 69. The mean of the data set is 50. The values 47 and 53 are removed, leaving 8 values. Which of the following is true of the new data set compared with the original? (SD means standard deviation.)

(A) Mean greater; SD smaller

(B) Mean unchanged; SD smaller

(C) Mean unchanged; SD larger ✓

(D) Mean smaller; SD larger

1. The sum of the original values is 500. The two removed values add to $47 + 53 = 100$, so the new sum is 400 and the new mean is $\frac{400}{8} = 50$: the mean is unchanged.

2. The removed values are 3 below and 3 above the mean, the two values closest to it.

3. The values left are farther from the mean on average. The sum of squared distances drops only from 1,266 to 1,248, but it is now shared among 8 values instead of 10: the variance goes from 126.6 to 156, so the SD is larger.

Answer: (C)

16. HARD · NO CALCULATOR NEEDED

A list of nine positive integers has a mean of 12 and a median of 9. What is the greatest possible value of the sum of the three largest integers in the list?

1. The nine integers add to $9 \times 12 = 108$. In order, the median is the fifth integer, so it is 9.

2. To make the three largest as big as possible, make the others as small as possible: the four smallest are at least 1 each, and the sixth integer must be at least the median, 9.

3. Everything else adds to $4(1) + 9 + 9 = 22$, so the three largest add to at most $108 - 22 = 86$.

4. This is reachable: $1, 1, 1, 1, 9, 9, 9, 68$ has sum 108 and median 9.

Answer: 86

17. HARD · NO CALCULATOR NEEDED

Seven positive integers, not necessarily distinct, have a sum of 101 and a median of 11. What is the greatest possible value of the mean of the three largest of these integers?

(A) 29 ✓ (B) 31.67

(C) 32.33 (D) 33.67

1. In order, the median of seven integers is the fourth one, so it is 11.
2. To make the three largest as big as possible, make the others as small as possible: the three smallest are 1 each, and the median is 11, so they add to $3 + 11 = 14$.
3. The three largest add to at most $101 - 14 = 87$. Their mean is $\frac{87}{3} = 29$.
4. This is reachable: $1, 1, 1, 11, 11, 11, 65$ has sum 101 and median 11.

Answer: (A)

18. HARD · NO CALCULATOR NEEDED

A data set has six values: 2, 3, 5, 12, 18, 20. Every value is increased by 4, and then a seventh value equal to the mean of the six increased values is added to the data set. Consider the following statements about the new data set of seven values compared with the original data set.
I. The mean is 4 greater.
II. The standard deviation is smaller.
III. The median is 4 greater.
Which of the statements must be true?

(A) I, II and III (B) I only

(C) I and II only ✓ (D) II and III only

1. Original mean: $\frac{60}{6} = 10$. Adding 4 to every value raises the mean to 14, and a new value equal to that mean leaves it at 14. I is true.
2. Adding 4 to every value does not change the spread. The new value sits exactly at the mean, so it adds nothing to the sum of squared distances but increases the number of values. The standard deviation is smaller, so II is true.
3. Original median: $\frac{5 + 12}{2} = 8.5$. The new set in order is $6, 7, 9, 14, 16, 22, 24$, so its median is 14, which is 5.5 more, not 4 more. III is false.

Answer: (C)

19. HARD · NO CALCULATOR NEEDED

A data set has a median of 52 and an interquartile range (IQR) of 14. Each value x in the data set is replaced by $15 - \frac{x}{2}$. What is the sum of the median and the IQR of the new data set?

(A) 3 (B) 11

(C) -18 **(D) -4 ✓**

1. The rule $x \to 15 - \frac{x}{2}$ is a straight-line change, so the median goes through the same rule: $15 - \frac{52}{2} = -11$.

2. Adding 15 does not change the spread, and multiplying by $-\frac{1}{2}$ halves it (the negative sign only flips the order). The IQR becomes $14 \times \frac{1}{2} = 7$.

3. Sum: $-11 + 7 = -4$.

Answer: (D)

20. HARD · NO CALCULATOR NEEDED

Let a, b, and c be positive integers. If the average (arithmetic mean) of a, b, and c is 100, which of the following is NOT a possible value of any of the three integers?

(A) 150 (B) 250

(C) 298 **(D) 299 ✓**

1. The mean is 100, so $a + b + c = 3 \times 100 = 300$.

2. One integer is largest when the other two are smallest. Each of them is at least 1, so the largest integer is at most $300 - 1 - 1 = 298$.

3. So 150, 250 and 298 are all possible values. 299 is not: it would force the other two to add to 1, but they add to at least 2.

Answer: (D)

21. HARD · NO CALCULATOR NEEDED

A researcher wants to determine whether a flipped-classroom method (Method A) causes higher unit-test scores than traditional lecture (Method B) among Grade 11 Algebra II students in a large school district. From all Grade 11 Algebra II sections in the district, the researcher randomly assigns 10 sections to use Method A and 10 sections to use Method B for one unit, then compares the sections' mean test scores. Which statement is best supported by this study design?

(A) Because sections were randomly assigned to a method, a causal conclusion about the effect of the method on test scores can be drawn for the Grade 11 Algebra II sections in this district, but the results should not be generalized beyond this district. ✓

(B) Because sections, rather than individual students, were randomly assigned, no causal conclusion can be drawn at all.

(C) Because an entire school district was involved, the results can be generalized to all high school students nationally.

(D) Because the researcher chose which sections received which method, this is an observational study, so no causal conclusions are valid.

1. The researcher assigned the sections to the two methods at random. Random assignment is what supports a cause-and-effect conclusion about the method for the sections studied.
2. The statements that call it an observational study or say that no causal conclusion can be drawn at all are wrong: random assignment of sections is still random assignment.
3. The sections come from this one district. Nothing was randomly selected from all districts in the country, so the results cannot be generalized nationally or beyond this district.

Answer: (A)

22. HARD · NO CALCULATOR NEEDED

For a research orchard, a scientist randomly selected 40 apple saplings from the saplings that were 3 years old at the orchard and randomly assigned them to two groups. Each group received a different irrigation schedule for 2 consecutive years to observe whether the schedule affects fruit yield. Based on the design of the study, what is the largest group to which these results can be applied?

(A) All 40 saplings that were selected at this orchard

(B) All apple saplings that were 3 years old at this orchard ✓

(C) All apple saplings that were 3 years old in the country

(D) All apple saplings that were 5 years old at this orchard

1. The saplings were randomly selected from the 3-year-old saplings at the orchard, so the results apply to that population.
2. Random assignment to the two irrigation schedules supports a cause-and-effect conclusion for that population.
3. The 5-year-old saplings and the saplings elsewhere in the country were not part of the selection, and the 40 saplings alone are a smaller group than the population. The largest group is all 3-year-old apple saplings at this orchard.

Answer: (B)

23. HARD · NO CALCULATOR NEEDED

At the end of a radio program, the host invited listeners to text in and respond to a poll asking, "Do you support extending school hours by 30 minutes?" By the end of the show, the host reported that 62% responded "Yes" and 35% responded "No." Which of the following best explains why the results are unlikely to represent the sentiments of all students' families in the country?

(A) The percentages do not add up to 100%, so any possible conclusions from the poll are invalid.

(B) Those who responded to the poll were not a random sample of all students' families in the country. ✓

(C) There were not 50% "Yes" responses and 50% "No" responses.

(D) The show did not allow listeners enough time to respond to the poll.

1. Listeners chose for themselves whether to text in. This is a voluntary-response sample.
2. A voluntary-response sample is not a random sample of all students' families, so the results are likely biased.
3. The percentages adding to 97% is not the problem (some listeners may not have answered the question), and the number of Yes and No answers or the time allowed does not make the sample representative.

Answer: (B)

24. HARD · NO CALCULATOR NEEDED

A polling agency surveyed 800 adults selected at random from a large city and asked, "Are you satisfied with the quality of public transportation in the city?" Of those surveyed, 71% responded that they were satisfied. Based on the results of the survey, which of the following statements must be true?

1. Of all adults in the city, 71% are satisfied with the quality of public transportation.
2. If another 800 adults selected at random from the city were surveyed, 71% of them would report they are satisfied.
3. If 800 adults selected at random from a different city were surveyed, 71% of them would report they are satisfied.

(A) None ✓

(B) 2 only

(C) 1 and 2 only

(D) 1 and 3 only

1. A sample percentage is only an estimate of the percentage for the whole city, so statement 1 does not have to be true.
2. A new random sample will not give exactly 71% again, because samples vary. Statement 2 does not have to be true.
3. Another city is a different population, so statement 3 does not have to be true either. None must be true.

Answer: (A)

25. HARD · NO CALCULATOR NEEDED

A survey of $n = 400$ randomly selected residents produces a margin of error of 4.5% at a given confidence level. Assuming the margin of error is inversely proportional to $\sqrt{n}$, what would the margin of error be, in percent, for a survey of $n = 1600$ residents at the same confidence level?

1. The margin of error is proportional to $\dfrac{1}{\sqrt{n}}$, so only the change in n matters.

2. The sample size goes from 400 to 1600, which is 4 times larger, and $\sqrt{4} = 2$.

3. The margin of error is divided by 2: $\dfrac{4.5}{2} = 2.25$ percent.

Answer: 2.25

26. HARD · NO CALCULATOR NEEDED

Based on a random sample survey, a researcher estimates that $24\% \pm 4\%$ of high school students in a district spend more than 2 hours a day on e-reading apps. A school administrator claims that 32% of the district's high school students spend more than 2 hours a day on such apps. Is the administrator's claim consistent with the survey's estimate?

(A) Yes, because 32% is within the range of 20% to 28% suggested by the estimate.

(B) Yes, because 32% is close to 24%.

(C) No, because the sample size is too small to draw any conclusion.

(D) No, because 32% is outside the range of 20% to 28% suggested by the estimate. ✓

1. The plausible values are $24 - 4 = 20$ to $24 + 4 = 28$ percent.

2. The administrator's 32% is above 28%, so it is outside that range.

3. So the claim is not consistent with the survey. The sample size is not the reason.

Answer: (D)

27. HARD · NO CALCULATOR NEEDED

Two polls use the same confidence level. For a poll of n people with sample proportion $\hat{p}$, the margin of error is proportional to $\sqrt{\dfrac{\hat{p}(1-\hat{p})}{n}}$. Poll A surveyed 625 people and had $\hat{p} = 0.50$. Poll B had $\hat{p} = 0.10$ and a margin of error half as large as the margin of error of Poll A. How many people did Poll B survey?

1. Poll A: $\sqrt{\dfrac{0.5(0.5)}{625}} = \sqrt{0.0004} = 0.02$.

2. Poll B has half of that: 0.01. So $\sqrt{\dfrac{0.1(0.9)}{n}} = 0.01$.

3. Square both sides: $\dfrac{0.09}{n} = 0.0001$, so $n = \dfrac{0.09}{0.0001} = 900$.

Answer: 900

28. HARD · NO CALCULATOR NEEDED

A researcher surveys a random sample of 625 workers in a city and calculates a 95% confidence interval of 24.2 to 27.8 minutes for the mean one-way commute time of all workers in the city. Assume a second random sample of 5,625 workers has similar variability and the same method is used. Which of the following statements must be true?

(A) About 95% of sampled commutes lie in the interval.

(B) The new margin is about 0.2 min.

(C) The population mean cannot be 28.1 min.

(D) The new margin is about 0.6 min. ✓

1. The interval 24.2 to 27.8 is centered at 26, so its margin of error is $\frac{27.8 - 24.2}{2} = 1.8$ minutes.
2. The new sample is $\frac{5625}{625} = 9$ times as large, and $\sqrt{9} = 3$, so the margin of error is divided by 3: $\frac{1.8}{3} = 0.6$ minutes.
3. The interval describes the mean of all workers, not individual commutes, and a 95% interval does not rule out values outside it. The new margin is the statement that must be true.

Answer: (D)

29. HARD · CALCULATOR

County P has 2,500 adult residents and county Q has 900. A random sample of county P residents found that 52% support a proposed bike lane, with a margin of error of 3 percentage points. A separate random sample of county Q residents found that 38% support it, with a margin of error of 4 percentage points. Based on the margins of error, what is the greatest possible value of (number of adult supporters in county P) − (number of adult supporters in county Q)?

(A) 1,033

(B) 1,069 ✓

(C) 525

(D) 958

1. The most supporters in county P come from the top of its range: $52 + 3 = 55\%$ of 2,500. Type `0.55*2500`: it returns 1,375.
2. The fewest supporters in county Q come from the bottom of its range: $38 - 4 = 34\%$ of 900. Type `0.34*900`: it returns 306.
3. The greatest difference is Type `1375-306`: it returns 1,069.

Answer: (B)

30. HARD · CALCULATOR

In a career-interest survey, 90 Grade 11 students intend a non-STEM track: 40 of them are male and 50 are female. Two of these students are selected at random, one after another, without replacement. What is the probability that both selected students are female? (Round to the nearest thousandth.)

1. First selection: $\frac{50}{90}$ chance of a female student. With no replacement, 49 of the remaining 89 students are female.
2. Multiply the two probabilities: Type `50/90*49/89`: it returns 0.3059.
3. Rounded to the nearest thousandth: 0.306.

Answer: 0.306

31. HARD · CALCULATOR

On January 1, 2024, there were 34.1 million mobile broadband subscriptions in a certain country. By January 1, 2025, the number of subscriptions had increased to 36.8 million. If a subscriber on January 1, 2025, is selected at random, the probability that the subscriber uses Carrier A is 0.45. There were p million Carrier A subscribers on January 1, 2025. To the nearest integer, what is the value of p?

1. The 2024 figure is not needed. Use the total on January 1, 2025: 36.8 million.

2. Carrier A has 45% of them: Type `36.8*0.45`: it returns 16.56.

3. To the nearest integer, $p = 17$.

Answer: 17

32. HARD · NO CALCULATOR NEEDED

The table below shows the distribution of employment type and sector for a group of employees at a company.

	Public sector	Private sector
Full-time	45	12
Part-time	9	x

If one of these employees who works in the private sector is chosen at random, the probability that the employee is part-time is $\frac{1}{5}$. What is the value of x?

1. The private-sector column has 12 full-time and x part-time employees, $12 + x$ in all.

2. The probability of part-time is $\frac{x}{12+x} = \frac{1}{5}$.

3. Cross-multiply: $5x = 12 + x$, so $4x = 12$ and $x = 3$.

Answer: 3

33. HARD · NO CALCULATOR NEEDED

	Online	Paper form
Robotics	60%	80%
Sports	25%	10%
Art	10%	5%
Music	5%	5%

A school activity survey asked each student to select the one activity they preferred for a school event. Some students responded online, and the others responded on a paper form. The table above shows the distribution of preferred activity, in percent, for each response type. For the survey, the number of paper-form responses was twice the number of online responses. If a student who preferred Art is selected at random, what is the probability that the student responded on the paper form? (Express your answer as a decimal or fraction, not as a percent.)

1. Pick numbers that fit: 100 online responses and 200 paper-form responses.

2. Art: 10% of 100 is 10 online, and 5% of 200 is 10 on paper.

3. Of the 20 Art responses, 10 were on paper: $\frac{10}{20} = \frac{1}{2}$.

Answer: $\frac{1}{2}$ **(or** 0.5**)**

34. HARD · NO CALCULATOR NEEDED

A conference has 300 participants. Each participant attended either the Technical track or the Business track, and either in person or online. Some of the data are shown.

	Technical	Business	Total
In person	70		
Online	50		
Total	120		300

Of the participants in the Business track, 35% attended online. How many of the 300 participants attended in person?

(A) 187 ✓ (B) 63

(C) 113 (D) 145

1. The Business track has $300 - 120 = 180$ participants.

2. Online Business: $0.35 \times 180 = 63$, so in-person Business is $180 - 63 = 117$.

3. In person in total: $70 + 117 = 187$.

Answer: (A)

35. HARD · DESMOS

A survey of 240 adults asked whether they commute by bicycle and whether they own a helmet. Of the adults who commute by bicycle, $\frac{3}{5}$ own a helmet. Of the adults who do not commute by bicycle, $\frac{1}{4}$ own a helmet. If one adult is selected at random from those who own a helmet, the probability that the selected adult commutes by bicycle is $\frac{12}{19}$. How many of the 240 adults own a helmet?

(A) 60 **(B)** 95 ✓

(C) 144 (D) 100

1. Let b be the number who commute by bicycle. Helmet owners: $\frac{3b}{5}$ among cyclists plus $\frac{240-b}{4}$ among the others.

2. The probability that a helmet owner is a cyclist is $\frac{3b/5}{3b/5 + (240-b)/4} = \frac{12}{19}$. Type `y=(3x/5)/(3x/5+(240-x)/4)` and `y=12/19` (letting x stand for b).

3. Tap the intersection: (100, 0.6316), so $b = 100$.

4. Helmet owners in all: Type `3*100/5+(240-100)/4`: it returns 95.

Answer: (B)

PART B · SELF-ASSESSMENT — 20 PROBLEMS

36. MEDIUM · CALCULATOR

The table shows the number of siblings reported by 20 students.

Number of siblings	0	1	2	3	4
Number of students	5	8	4	2	1

What is the mean number of siblings for these 20 students?

(A) 1.1 (B) 1.2

(C) 1.3 ✓ (D) 1.5

1. Each number of siblings is counted as many times as there are students: $0 \cdot 5 + 1 \cdot 8 + 2 \cdot 4 + 3 \cdot 2 + 4 \cdot 1$. Type `0*5+1*8+2*4+3*2+4*1` : it returns 26.
2. Divide by the 20 students. Type `26/20` : it returns 1.3.
3. The mean is 1.3 siblings.

Answer: (C)

37. MEDIUM · CALCULATOR

The table below shows the distribution of ages of 20 tutors enrolled in a training program.

Age	Frequency
20	7
21	5
22	3
23	2
24	1
30	2

Which of the following gives the correct order of the mean, median, and mode of the ages?

(A) mode < median < mean ✓ (B) mode < mean < median

(C) median < mode < mean (D) mean < mode < median

1. The mode is the most frequent age: 20, which appears 7 times.
2. There are 20 values, so the median is the average of the 10th and 11th. The running counts are 7 (age 20) and 12 (through age 21), so both are 21 and the median is 21.
3. Mean: Type `(20*7+21*5+22*3+23*2+24*1+30*2)/20` : it returns 22.05.
4. The two ages of 30 pull the mean above the median, so mode < median < mean.

Answer: (A)

38. Medium · Calculator

Data set A and data set B each consist of 20 values. The table shows the frequencies of the values for each data set. Which of the following statements best compares the means of the two data sets?

Value	Data set A frequency	Data set B frequency
30	2	7
35	5	6
40	6	4
45	7	3

(A) The mean of data set A is greater than the mean of data set B. ✓

(B) The mean of data set A is less than the mean of data set B.

(C) The mean of data set A is equal to the mean of data set B.

(D) There is not enough information to compare the means of the data sets.

1. Data set A: Type `(30*2+35*5+40*6+45*7)/20`: it returns 39.5.

2. Data set B: Type `(30*7+35*6+40*4+45*3)/20`: it returns 35.75.

3. Since $39.5 > 35.75$, the mean of A is greater. The tables give every value, so there is enough information.

Answer: (A)

39. Medium · No calculator needed

The mean of 5 numbers is 14. One of the numbers, 20, is replaced by 8. What is the mean of the new set of 5 numbers? (Enter as a decimal.)

1. Original sum: $5 \times 14 = 70$.

2. Replacing 20 with 8 lowers the sum by 12: $70 - 20 + 8 = 58$.

3. New mean: $\frac{58}{5} = 11.6$.

Answer: 11.6

40. Medium · No calculator needed

If the average (arithmetic mean) of w and x is 40, the average of p and q is 40, and the average of m and n is 40, what is the average of w, x, p, q, m, and n?

1. Each pair has mean 40, so each pair adds to $2 \times 40 = 80$.

2. The six numbers add to $3 \times 80 = 240$.

3. Their mean is $\frac{240}{6} = 40$.

Answer: 40

41. Medium · No calculator needed

A researcher surveyed a random sample of chemistry teachers selected from all chemistry teachers in a certain state. The teachers were asked to name the textbook publisher they currently use. What is the largest population to which the results of the survey can be generalized?

(A) All teachers in the country

(B) All chemistry teachers in the country

(C) All chemistry teachers in the state ✓

(D) All teachers in the state

1. Results of a random sample can be generalized only to the population the sample was drawn from.
2. The sample was drawn from chemistry teachers in one state, so the population is all chemistry teachers in that state.
3. Teachers of all subjects, or chemistry teachers in the whole country, were never available to be selected.

Answer: (C)

42. Medium · No calculator needed

A random sample of 150 residents of a city was surveyed about whether they support a new recycling program. All 150 responded, and 64% said they support it. Based on this information, which of the following statements must be true?

I. Exactly 96 of the 150 residents surveyed said they support the recycling program.

II. Of all city residents, exactly 64% support the recycling program.

(A) Neither

(B) I only ✓

(C) II only

(D) I and II

1. All 150 residents responded, so 64% of 150 is a count of people in the sample: $0.64 \times 150 = 96$ residents said they support the program. Statement I must be true.
2. Statement II is about all city residents. A sample percentage is only an estimate of the population percentage, so it does not have to equal it exactly.
3. Only statement I must be true.

Answer: (B)

43. Medium · No calculator needed

A random sample of 500 residents in a county found that 62% support a proposed residential zoning initiative, with a margin of error of 4 percentage points at the 95% confidence level. Which of the following is the most appropriate interpretation of this result?

(A) Exactly 62% of all residents in the county support the initiative.

(B) It is plausible that between 58% and 66% of all residents in the county support the initiative. ✓

(C) Exactly 95% of residents in the county support the initiative.

(D) The survey proves beyond doubt that a majority of residents support the initiative.

1. The margin of error is 4 percentage points, so the plausible values are $62 - 4 = 58$ to $62 + 4 = 66$ percent.
2. A survey gives an estimate with a range of plausible values, not the exact value, and it cannot prove anything beyond doubt.
3. The 95% is the confidence level, not the percentage of residents who support the initiative.

Answer: (B)

44. Medium · No calculator needed

Researchers studying a reforestation program selected a random sample of 250 oak trees planted through the program and measured each tree's growth rate. Based on this sample, which of the following is the most appropriate conclusion?

(A) The average growth rate of the sampled trees can be used to estimate the average growth rate of all oak trees planted through the reforestation program. ✓

(B) The average growth rate of the sampled trees can be used to estimate the average growth rate of all trees, of any species, in the country.

(C) The average growth rate of the sampled trees is exactly equal to the average growth rate of all oak trees planted through the reforestation program.

(D) The average growth rate of the sampled trees cannot be used to estimate the average growth rate of any other trees.

1. Results of a random sample apply to the population it was drawn from: here, the oak trees planted through this program.
2. So the sample average can be used as an estimate for those trees, but it is only an estimate, not an exact value.
3. It cannot speak for trees of every species in the country, and it is not exactly equal to the program's average. Only the statement about estimating the average for the program's oak trees is correct.

Answer: (A)

45. Medium · No calculator needed

A survey of a random sample of employees at a company estimates that the mean daily commute time is 34 minutes, with an associated margin of error of 4 minutes. Which of the following is the most plausible value for the actual mean commute time, in minutes, of all employees at the company?

(A) 25

(B) 31 ✓

(C) 39

(D) 44

1. The plausible values run from $34 - 4 = 30$ to $34 + 4 = 38$ minutes.
2. Check each option: 25 is below 30, 39 and 44 are above 38.
3. Only 31 lies in the range.

Answer: (B)

46. Medium · No calculator needed

A poll of a random sample of residents in a city district estimates that 42% of all residents support a new public transit expansion, with a margin of error of 3.5 percentage points. Which of the following is the most plausible value for the actual percentage of all residents in the district who support the expansion?

(A) 36%

(B) 41% ✓

(C) 47%

(D) 50%

1. The plausible values run from $42 - 3.5 = 38.5$ to $42 + 3.5 = 45.5$ percent.
2. Check each option: 36% is below 38.5%, and 47% and 50% are above 45.5%.
3. Only 41% lies in the range.

Answer: (B)

47. Medium · Calculator

A survey of a random sample of students at a university estimates that $22\% \pm 3\%$ of the university's 8,000 students use the campus shuttle daily. Based on this estimate, which of the following is closest to a plausible range for the number of students at the university who use the shuttle daily?

(A) $1{,}400 < n < 1{,}800$

(B) $1{,}520 < n < 2{,}000$ ✓

(C) $1{,}760 < n < 1{,}840$

(D) $1{,}600 < n < 2{,}200$

1. The margin of error is 3 points, so the plausible percentages are $22 - 3 = 19$ to $22 + 3 = 25$ percent.

2. Lower end: Type `0.19*8000`: it returns 1,520. Upper end: Type `0.25*8000`: it returns 2,000.

3. The plausible range is $1{,}520 < n < 2{,}000$.

Answer: (B)

48. Medium · No calculator needed

In a random sample of 60 laptops tested at a distribution center, 15 were found to have a manufacturing defect. Based on this sample, which of the following is the best conclusion about the number of defective laptops in a shipment of 2,400 laptops from the same production run?

(A) Exactly 600 laptops in the shipment are defective.

(B) At least 600 laptops in the shipment are defective.

(C) A best estimate is that 600 laptops in the shipment are defective. ✓

(D) At most 600 laptops in the shipment are defective.

1. The sample defect rate is $\frac{15}{60} = 0.25$, which is 25%.

2. Applying the rate to the shipment: $0.25 \times 2400 = 600$.

3. This came from a sample, so it is a best estimate, not an exact count, a minimum or a maximum.

Answer: (C)

49. Medium · Calculator

In a random sample of 500 registered voters surveyed about a proposed municipal budget, 310 said they support it. There are 45,000 registered voters in total. Based on the sample, what is the expected number of registered voters who support the proposed budget?

1. Sample proportion in favor: Type `310/500`: it returns 0.62.

2. Apply it to all 45,000 registered voters: Type `0.62*45000`: it returns 27,900.

3. The expected number is 27,900 voters.

Answer: 27,900

50. Medium · No calculator needed

In a career-interest study, 200 Grade 11 students were surveyed on their intended career track.

	STEM track	Non-STEM track	Total
Male	60	40	100
Female	50	50	100
Total	110	90	200

If a student is selected at random from those surveyed and the student is female, what is the probability that the student intends a STEM track?

(A) 0.45

(B) 0.50 ✓

(C) 0.55

(D) 0.60

1. Restrict to the condition: only the 100 female students count.

2. Of those, 50 intend a STEM track.

3. $P = \dfrac{50}{100} = 0.5$.

Answer: (B)

51. Medium · No calculator needed

The table below shows the number of students in two grade levels at a school who do and do not own a graphing calculator.

	Owns a graphing calculator	Does not own a graphing calculator	Total
Grade 10	18	7	25
Grade 11	22	3	25
Total	40	10	50

If one of these 50 students is selected at random, what is the probability that the student is in Grade 11 and does not own a graphing calculator?

(A) $\dfrac{3}{50}$ ✓

(B) $\dfrac{7}{50}$

(C) $\dfrac{10}{50}$

(D) $\dfrac{22}{50}$

1. Read the row for Grade 11 and the column for does not own a graphing calculator.

2. That cell holds 3 students, and the total number of students is 50.

3. $P = \dfrac{3}{50}$.

Answer: (A)

52. MEDIUM · NO CALCULATOR NEEDED

The table below shows the distribution of 50 students by club membership.

	Coding club	Not coding club	Total
Robotics club	15	8	23
Not robotics club	10	17	27
Total	25	25	50

If one of these 50 students is selected at random, what is the probability that the student is in the coding club but not the robotics club? (Express your answer as a decimal or fraction, not as a percent.)

1. Read the row for not robotics club and the column for coding club.

2. That cell holds 10 students, and the total number of students is 50.

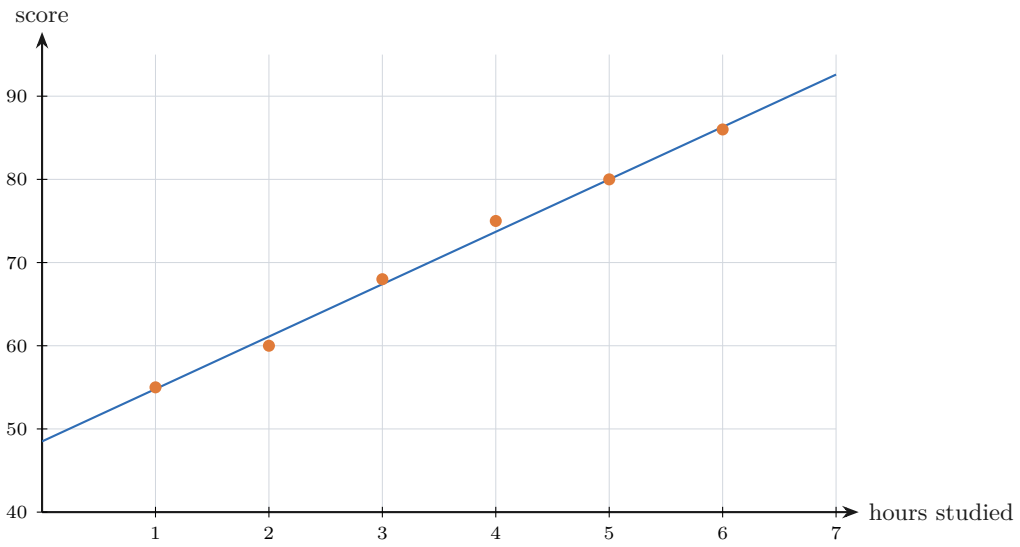
3. $P = \dfrac{10}{50} = \dfrac{1}{5} = 0.2$.

Answer: $\dfrac{1}{5}$ **(or** 0.2**)**

53. MEDIUM · NO CALCULATOR NEEDED

A teacher records hours studied (x) and exam score (y) for 6 students.

Hours studied (x)	1	2	3	4	5	6
Exam score (y)	55	60	68	75	80	86



The line of best fit for this data is approximately $y = 6.3x + 48.5$. Which of the following is the best interpretation of the slope of this line?

(A) The predicted exam score for a student who studies 0 hours is 6.3.

(B) Each additional hour studied is associated with a predicted increase of about 6.3 points on the exam. ✓

(C) Each additional point on the exam is associated with a predicted increase of about 6.3 hours studied.

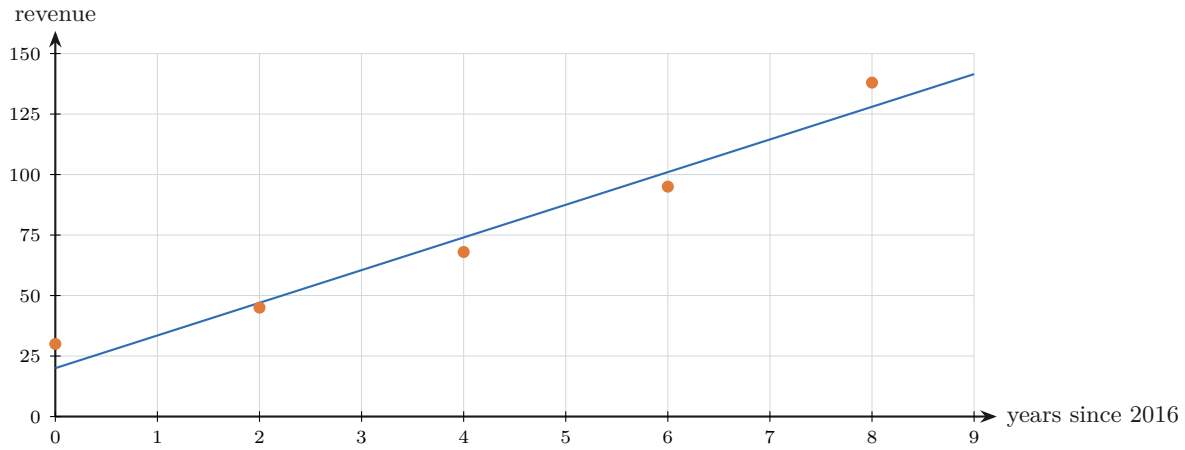
(D) The exam score increases by exactly 6.3 points for every student, regardless of hours studied.

1. In $y = mx + b$ the slope m is the change in y for each increase of 1 in x.
2. Here y is the exam score and x is the hours studied, so 6.3 is the predicted change in score for each extra hour.
3. The y-intercept (48.5) is the predicted score at 0 hours, and a best-fit line gives predictions, not exact changes for every student.

Answer: (B)

54. HARD · NO CALCULATOR NEEDED

A company's annual online-sales revenue, in millions of dollars, is recorded for years since 2016 (x). The approximate data points are $(0, 30)$, $(2, 45)$, $(4, 68)$, $(6, 95)$ and $(8, 138)$. The scatterplot shows the data and a line used to model it, $y = 13.5x + 20$.



What is the residual for the data point at $x = 8$?

(A) -10

(B) 0

(C) 10 ✓

(D) 128

1. The line predicts, at $x = 8$: $13.5(8) + 20 = 128$.

2. Residual = actual − predicted $= 138 - 128 = 10$.

3. The actual point is above the line, so the residual is positive. (-10 comes from subtracting in the wrong order.)

Answer: (C)

55. HARD · NO CALCULATOR NEEDED

A scatterplot showing the relationship between daily study time (x, in hours) and nightly sleep time (y, in hours) for a group of students has a correlation coefficient of $r = -0.82$. Which of the following best describes this relationship?

(A) A strong, positive linear relationship between study time and sleep time.

(B) A strong, negative linear relationship between study time and sleep time. ✓

(C) A weak, negative linear relationship between study time and sleep time.

(D) No linear relationship between study time and sleep time.

1. The sign of r gives the direction: $r = -0.82$ is negative, so as study time goes up, sleep time tends to go down.

2. The size of r gives the strength: 0.82 is close to 1, so the relationship is strong.

3. So it is a strong, negative linear relationship.

Answer: (B)